

# S2S-04: OneAgent and ActiveGate Cutover

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**Series:** S2S | **Notebook:** 4 of 12 | **Created:** March 2026 | **Last Updated:** 03/23/2026

## Overview

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Migrating OneAgents and ActiveGates is the most operationally visible part of a SaaS-to-SaaS migration. Hosts must be reconfigured to report to the target tenant, which requires coordination with application teams, change management, and a clear rollback plan.

**Operator Version:** This notebook targets Dynatrace Operator **v1.8.1** (February 2026) with DynaKube API **v1beta5/v1beta6**. Operator 1.8.x removes v1beta3 and auto-converts to v1beta6. Both v1beta5 and v1beta6 are supported; v1beta6 adds OTLP exporter configuration.

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# Prerequisites

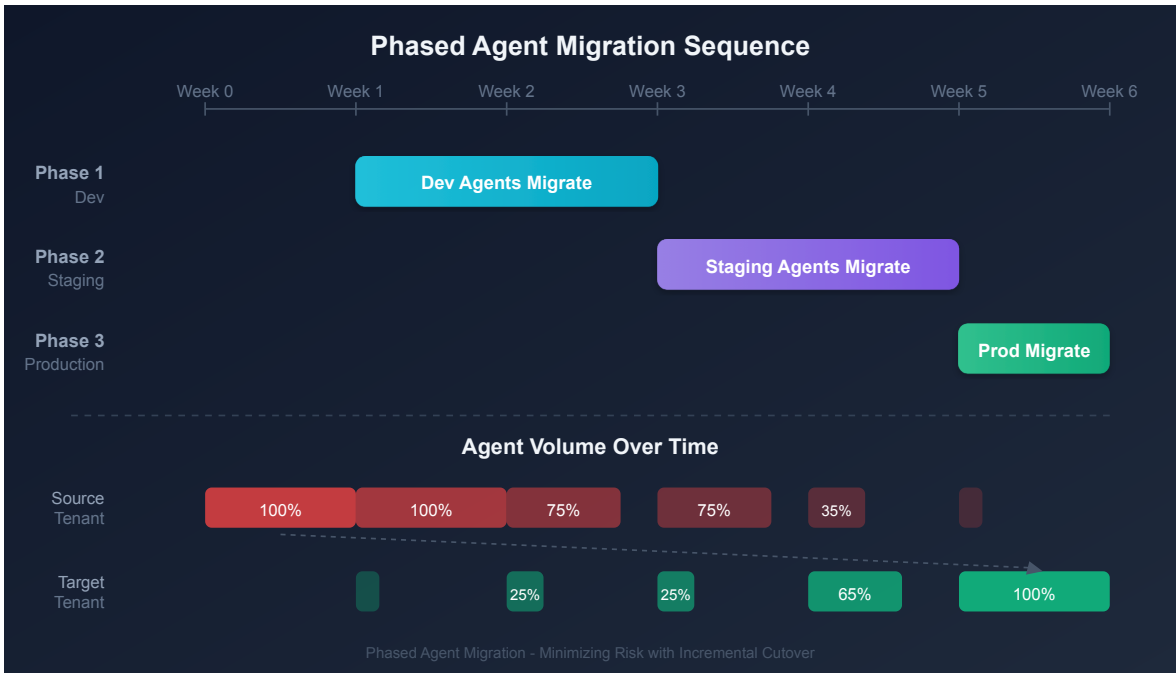
| Requirement        | Details                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------|
| Target Tenant      | Provisioned with OneAgent deployment configured                                                |
| Dynatrace Operator | v1.8.1+ (Helm chart <code>oci://public.ecr.aws/dynatrace/dynatrace-operator</code> )           |
| DynaKube API       | <code>dynatrace.com/v1beta5</code> or <code>v1beta6</code> (v1beta3 removed in Operator 1.8.0) |
| Change Management  | Approved change window for agent cutover                                                       |
| Network            | Target tenant URL reachable from all hosts                                                     |
| Tokens             | PaaS token or installer download token for target tenant                                       |

## 1. Agent Migration Strategies

### Approach Comparison

| Strategy              | Description                                            | Risk                                    | Downtime                |
|-----------------------|--------------------------------------------------------|-----------------------------------------|-------------------------|
| Big Bang              | All agents switch at once                              | High — no partial rollback              | Minutes (config change) |
| Phased by Host Group  | Migrate host groups one at a time                      | Low — rollback per group                | None (rolling)          |
| Phased by Environment | Dev → Staging → Production                             | Lowest — full validation between phases | None (rolling)          |
| Blue-Green            | New agents in target, old agents in source, DNS switch | Lowest — instant rollback               | None                    |

**Recommendation:** Phased by environment. Migrate dev first, validate for 1-2 weeks, then staging, then production. This gives Davis AI time to establish baselines in the target tenant.



## 2. OneAgent Reconfiguration

### Using `oneagentctl`

The `oneagentctl` command-line tool reconfigures a running OneAgent to report to a different tenant:

```
# Linux – reconfigure to target tenant
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl \
  --set-server=https://&lt;target-tenant>.live.dynatrace.com/communication
\
  --set-tenant=&lt;target-tenant-id> \
  --set-tenant-token=&lt;target-paas-token>;

# Windows – same operation
"C:\ProgramData\dynatrace\oneagent\agent\tools\oneagentctl.exe" ^
  --set-server=https://&lt;target-tenant>.live.dynatrace.com/communication
^
  --set-tenant=&lt;target-tenant-id> ^
  --set-tenant-token=&lt;target-paas-token>;

# Verify the change
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --get-server
```

**Note:** The OneAgent restarts its communication after `oneagentctl` changes. Monitored applications are NOT restarted — there is no application downtime.

## Automation at Scale

For large fleets, automate with your configuration management tool:

| Tool                       | Approach                                                                   |
|----------------------------|----------------------------------------------------------------------------|
| <b>Ansible</b>             | Playbook with <code>command</code> module running <code>oneagentctl</code> |
| <b>Chef</b>                | Recipe updating OneAgent configuration                                     |
| <b>Puppet</b>              | Manifest managing OneAgent settings                                        |
| <b>AWS Systems Manager</b> | Run Command document across EC2 fleet                                      |
| <b>Azure Automation</b>    | Runbook executing on VM fleet                                              |
| <b>GCP OS Config</b>       | Policy applying <code>oneagentctl</code> changes                           |

## 3. ActiveGate Migration

ActiveGates must be reinstalled — they cannot be reconfigured like OneAgents:

| Step | Action                                  | Notes                                    |
|------|-----------------------------------------|------------------------------------------|
| 1    | Install new ActiveGate in target tenant | Download installer from target tenant UI |
| 2    | Configure network zones (if used)       | Match source network zone topology       |
| 3    | Configure ActiveGate routing            | Point OneAgents to new ActiveGate        |
| 4    | Verify connectivity                     | Check ActiveGate health in target UI     |
| 5    | Decommission source ActiveGate          | After all OneAgents migrated             |

## ActiveGate Types to Migrate

| Type                          | Migration Path                   | Notes                                              |
|-------------------------------|----------------------------------|----------------------------------------------------|
| <b>Environment ActiveGate</b> | Install new, reconfigure routing | OneAgents auto-discover new AG                     |
| <b>Cluster ActiveGate</b>     | Not applicable to SaaS           | SaaS does not use Cluster AGs                      |
| <b>Synthetic ActiveGate</b>   | Install new, reassign locations  | Synthetic monitors must update location references |

## 4. Kubernetes Operator Migration

### Helm Installation for Target Tenant

```
# Install Operator v1.8.1 in target cluster
helm upgrade dynatrace-operator oci://public.ecr.aws/dynatrace/dynatrace-operator \
  --version 1.8.1 \
  --create-namespace --namespace dynatrace \
  --install \
  --atomic
```

### DynaKube CR Update

Update the DynaKube custom resource to point to the target tenant:

```
apiVersion: dynatrace.com/v1beta5
kind: DynaKube
metadata:
  name: dynakube
  namespace: dynatrace
spec:
  apiUrl: https://<target-tenant>.live.dynatrace.com/api
# ... rest of configuration unchanged
```

**API Version Warning:** Operator 1.8.x removes `v1beta3` and auto-converts existing CRDs to `v1beta6`. Both `v1beta5` and `v1beta6` are supported. If your manifests use `v1beta3`, update to `v1beta5` or `v1beta6` before upgrading. The Operator emits Kubernetes warning events if the CRD version does not match.

## Migration Steps

| Step | Action                                                                   | Tool                    |
|------|--------------------------------------------------------------------------|-------------------------|
| 1    | Create <code>dynatrace</code> namespace secret with target tenant tokens | kubectl                 |
| 2    | Update DynaKube CR <code>apiUrl</code> to target tenant                  | kubectl / ArgoCD        |
| 3    | Restart Dynatrace Operator                                               | kubectl rollout restart |
| 4    | Verify pods reconnect to target                                          | Target tenant UI        |
| 5    | Verify enrichment tags populate                                          | DQL query in target     |

## Cloud-Specific K8s Considerations

| Platform       | K8s Service   | Key Differences                                                    |
|----------------|---------------|--------------------------------------------------------------------|
| <b>AWS</b>     | EKS           | IAM roles for service accounts (IRSA) must reference target tenant |
| <b>Azure</b>   | AKS           | Managed Identity or Service Principal for target tenant            |
| <b>GCP</b>     | GKE           | Workload Identity mapping for target tenant                        |
| <b>On-Prem</b> | K8s/OpenShift | Network routing to target tenant URL                               |

## 5. Dynatrace Operator Resource Overhead

When planning capacity for the target cluster, account for the resource overhead introduced by the Dynatrace Operator and its components. These values are from the [Operator v1.8.1 Helm chart defaults](#).

### Control Plane Components (per cluster)

| Component                  | Replicas | CPU Request | CPU Limit   | Memory Request | Memory Limit |
|----------------------------|----------|-------------|-------------|----------------|--------------|
| <b>Operator</b>            | 1        | 50m         | 100m        | 64Mi           | 128Mi        |
| <b>Webhook</b>             | 2        | 300m each   | 300m each   | 128Mi each     | 128Mi each   |
| <b>Control plane total</b> | —        | <b>650m</b> | <b>700m</b> | <b>320Mi</b>   | <b>384Mi</b> |

**Webhook HA:** Operator 1.8.x defaults to 2 webhook replicas with topology spread constraints across zones and nodes, plus a PodDisruptionBudget ( `minAvailable: 1` ).

## Per-Node DaemonSet Components

| Component                  | CPU Request | CPU Limit | Memory Request | Memory Limit |
|----------------------------|-------------|-----------|----------------|--------------|
| OneAgent                   | 100m        | 300m      | 512Mi          | 1.5Gi        |
| CSI Driver (server)        | 50m         | 50m       | 100Mi          | 100Mi        |
| CSI Driver (provisioner)   | 300m        | —         | 100Mi          | —            |
| CSI Driver (registrar)     | 20m         | 20m       | 30Mi           | 30Mi         |
| CSI Driver (livenessprobe) | 20m         | 20m       | 30Mi           | 30Mi         |
| CSI Driver (init)          | 50m         | 50m       | 100Mi          | 100Mi        |
| Per-node total             | ~540m       | ~440m+    | ~872Mi         | ~1.76Gi      |

## Total Cluster Overhead by Size

| Cluster Size | Control Plane | Node Overhead | Total CPU Request | Total Memory Request |
|--------------|---------------|---------------|-------------------|----------------------|
| 5 nodes      | 650m          | 2,700m        | ~3.4 cores        | ~4.7 Gi              |
| 10 nodes     | 650m          | 5,400m        | ~6.1 cores        | ~9.0 Gi              |
| 20 nodes     | 650m          | 10,800m       | ~11.5 cores       | ~17.8 Gi             |
| 50 nodes     | 650m          | 27,000m       | ~27.7 cores       | ~44.0 Gi             |
| 100 nodes    | 650m          | 54,000m       | ~54.7 cores       | ~87.5 Gi             |

**In practice:** OneAgent typically consumes 50-80m CPU per node under normal load. The 100m request provides headroom for code injection and deep monitoring. For a typical 8-vCPU node, Dynatrace uses about **6-7% of CPU** and **5-6% of memory** in request terms.



## ActiveGate Sizing (if deployed in-cluster)

| Cluster Size | Recommended Replicas | CPU Request (each) | Memory Request (each) |
|--------------|----------------------|--------------------|-----------------------|
| 1-10 nodes   | 1                    | 500m               | 1Gi                   |
| 11-30 nodes  | 2                    | 500m               | 1Gi                   |
| 31-50 nodes  | 2-3                  | 500m               | 1.5Gi                 |
| 50+ nodes    | 3+                   | 1000m              | 2Gi                   |

## Operator RBAC Roles (v1.8.1)

The Operator Helm chart creates RBAC roles for these components:

| RBAC Role                         | Purpose                                           | Created by Default |
|-----------------------------------|---------------------------------------------------|--------------------|
| <code>activeGate</code>           | ActiveGate deployment management                  | Yes                |
| <code>kubernetesMonitoring</code> | K8s API access for metrics                        | Yes                |
| <code>edgeConnect</code>          | Edge Connect management                           | Yes                |
| <code>extensions</code>           | Extensions 2.0 (includes Prometheus and database) | Yes                |
| <code>telemetryIngest</code>      | OpenTelemetry/metric ingest                       | Yes                |
| <code>logMonitoring</code>        | Log monitoring access                             | Yes                |
| <code>oneAgent</code>             | OneAgent DaemonSet management                     | Yes                |
| <code>kspm</code>                 | Kubernetes Security Posture Management            | Yes                |

## 6. Network Zone Reconfiguration

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If the source tenant uses network zones, recreate them in the target:

```
# Export network zone configuration from source
monaco download manifest.yaml --environment source \
  --specific-settings builtin:networkzones

# Apply to target
monaco deploy manifest.yaml --environment target
```

## Network Zone Mapping

| Source Zone  | Target Zone  | Hosts | Notes                                   |
|--------------|--------------|-------|-----------------------------------------|
| zone-us-east | zone-us-east | 50    | Same name, new tenant                   |
| zone-eu-west | zone-eu-west | 30    | Same name, new tenant                   |
| zone-dmz     | zone-dmz     | 10    | ActiveGate routing must be reconfigured |

## 7. Rollback Procedures

| Scenario               | Rollback Action                                            | Impact                          |
|------------------------|------------------------------------------------------------|---------------------------------|
| OneAgent not reporting | Run <code>oneagentctl</code> with source tenant parameters | 1-2 minutes per host            |
| DynaKube cutover fails | Revert DynaKube CR to source <code>apiUrl</code>           | Pods reconnect within minutes   |
| ActiveGate unhealthy   | Point OneAgents back to source AG                          | Requires AG to still be running |
| Mass rollback needed   | Ansible/SSM playbook with source parameters                | Fleet-wide in minutes           |

**Critical:** Keep the source tenant running throughout the migration window. Do not decommission source ActiveGates or revoke source tokens until all agents are confirmed in the target.

## Next Steps

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Continue to **S2S-05: Settings and Configuration Import** to apply exported configuration to the target tenant.

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